8. Depth-First Search (DFS)

Aim

To implement Depth-First Search to traverse a graph by exploring one branch deeply before backtracking.

Algorithm

1. Create the graph using an adjacency list.
2. Start from node A and create an empty visited set.
3. If the node has not been visited, mark it visited and print it.
4. Recursively visit each unvisited neighbour.
5. Stop when all reachable nodes have been visited.

Program

*AI.py - C:/Users/Barath/AppData/Local/Programs/

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```
graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": [],
    "F": []
}

def dfs(node, visited):
    if node in visited:
        return
    visited.add(node)
    print(node, end=" ")
    for neighbour in graph[node]:
        dfs(neighbour, visited)

dfs("A", set())
```

Output

```
==========
A B D E C F
```

Result

Thus, DFS traversed all reachable nodes of the graph from node A.